N-Digit Numbers with Increasing Digits

Difficulty: **Medium** Accuracy: **49.66%** Submissions: **20K+** Points: **4**

Given an integer **n**, return all the **n** digit numbers in increasing order, such that their digits are in strictly increasing order(from left to right).

Examples :

Input: n = 1
Output: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
Explanation: Single digit numbers are considered to be strictly increasing order.

Input: n = 2
Output: [12, 13, 14, 15, 16, 17, 18, 19, 23....79, 89]
Explanation: For n = 2, the correct sequence is 12 13 14 15 16 17 18 19 23 and so on up to 89.

Input: n = 15
Output: []
Explanation: No such number exist.

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```python
class Solution:
    def increasingNumbers(self, n):
        if n == 1:
            return [i for i in range(10)]

        if n > 9:
            return []

        ans = []

        def dfs(start, length, num):
            if length == n:
                ans.append(num)
                return

            for digit in range(start, 10):
                dfs(digit + 1, length + 1, num * 10 + digit)

        for first_digit in range(1, 10):
            dfs(first_digit + 1, 1, first_digit)

        return ans
```